



UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 2

**SYSTEM ANALYSIS AND DESIGN**

**DR.Aryati**

| NO | NAME                                     | MATRIC NO |
|----|------------------------------------------|-----------|
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## Question 1

- a. The system's feasibility can be assessed by evaluating whether the system can be built and maintained with the available resources, skills, and technology.
- b. In terms of operational feasibility, it is to see whether the user would use the new system or not. Based on Abu's interview, he found out that the users, managers, and people with contact with the system approve of the new system. This goes to show that the new system will be used when it is implemented.
- c. In terms of economic feasibility, the new system change is a considerable investment hence showing the uncertainty of economics for the new system change. For example, they have to invest their money in the systems analysts' time, cost of systems study, cost of employees salary, cost of hardware, and cost of system development.
- d. Regarding technological feasibility, even though the existing technology is enough, I believe it is preferable to advance to a newer version. As time goes on, technical system upgrades result in reduced operating costs and increased efficiency. Furthermore, this will make maintenance smoother.
- e. So, the conclusion is we do not recommend changing to a full-blown system right away but testing the new system by using the modular conversion where the new system can be implemented in a certain place first. This way they can have a smooth transition to the new system and also reduce the risk of system failure

## Question 2

| no. | Task                                                                                                                                                                               | Duration (week) | predecessor |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------|
| 1   | Planning phase :<br>1.1 define problem statement<br>1.2 determine the objective<br>1.3 analyze current system and trend                                                            | 1 week          |             |
| 2   | Analysation phase :<br>2.1 make the proposal<br>2.2 identify the risk of the system<br>2.3 find the possible solution<br>2.4 initialize the cost-benefit                           | 4 week          | 1           |
| 3   | Proposal submission :<br>3.1 update the proposal<br>3.2 submit the proposal                                                                                                        | 1 day           | 1 , 2       |
| 4   | Decision Making :<br>4.1 review proposal<br>4.2 make decision on proposal acceptance                                                                                               | 1 week          | 3           |
| 5   | Financial planning :<br>5.1 obtain budget for the system                                                                                                                           | 2 week          | 4           |
| 6   | Development Phase :<br>6.1 system development<br>6.1.1 backend development<br>6.1.2 frontend development<br>6.2 testing and debugging system<br>6.3 Combining with Current Systems | 14 week         | 4 , 5       |
| 7   | Implementation phase :<br>7.1 training for staff<br>7.2 deployed the system<br>7.3 monitor the feedback                                                                            | 5 week          | 6           |
| 8   | Maintenance and support :<br>8.1 Determine Upkeep Protocols<br>8.2 Provide Ongoing Support<br>8.3 Updates and improvements on a regular basis                                      | On going        | 6           |

3.

Title:

COMPARISON BETWEEN 2 DIFFERENT METHODOLOGY FACED BY MANAGERS WHILE MANAGING A TEAM MEMBERS THAT IS SCATTERED AROUND THE WORLD

Introduction :

There are a few methodologies that have been created to develop a system. There are two that are most commonly used which are waterfall methodology that is more traditional and agile methodology that is most of the company often used. These two methodologies have their own key differences especially as a manager that has to manage a team that is scattered around the world.

Key differences :

- Flexibility and adaptability

Agile methodology specializes in a flexible and adaptable approach. This means adjustments can be made at any point during the project. This is beneficial in team management as communication and coordination can be challenging due to time zone differences. Waterfall methodology is more rigid in comparison, with the emphasis on planning and predictability. Once the plan is set, it is hard to make adjustments afterwards especially when meetings cannot be held.

- Team Functionality

Agile methodology requires team members that have a variety of skills that are related to that project. This way, they can work on different phases of the project without any problems. For example, if one member that is responsible for a particular phase is sick, other members can serve as a backup. This prevents the project from being delayed. Meanwhile, using the traditional methodology, the team members are created for that specific task only. So they are more limited compared to agile methodology. The manager might experience project delays because of a lack of members to carry on the specific task when one of them is on leave.

- Communication

Agile methodology uses informal communication. This way they can interact regularly with each member and know the progress of the project in each phase. While traditional methodology uses a formal way of communication where they need to discuss each phase thoroughly before transitioning to a new one. This consumes a lot of time and the managers will have a hard time to find a slot where everyone is available to make a meeting.

Conclusion :

In conclusion, both of the methodologies have their pros and cons. As a manager, they need to choose an effective methodology to manage the project to avoid as many problems as possible, especially while handling teams that live in different time zones.

4.

A.

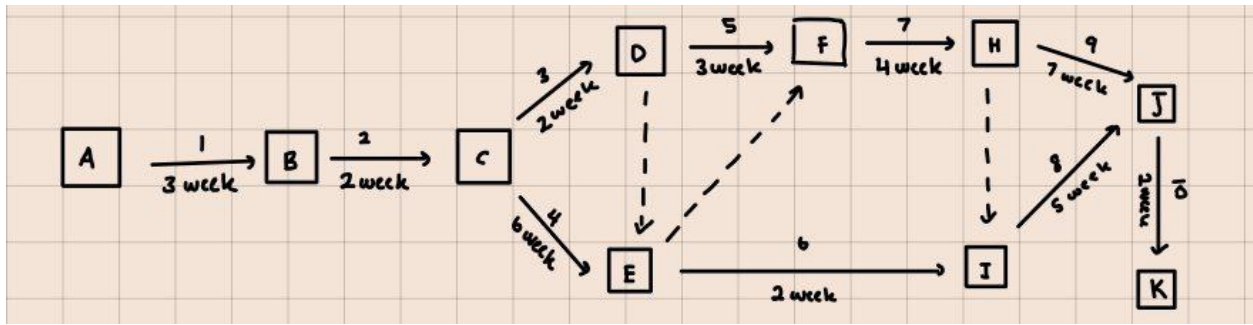
| costs             | Year 0 | Year 1  | Year 2  | Year 3  |  |
|-------------------|--------|---------|---------|---------|--|
| Development costs |        |         |         |         |  |
| ▪ Hardware        | 50 000 |         |         |         |  |
| ▪ Software        | 15 000 |         |         |         |  |
| ▪ Consulting      | 80 000 |         |         |         |  |
| ▪ Training        | 30 000 |         |         |         |  |
| ▪ Data Conversion | 40 000 |         |         |         |  |
| Total             | 215000 |         |         |         |  |
| Production costs  |        |         |         |         |  |
| ▪ Supplies        |        | 15 000  | 15 000  | 15 000  |  |
| ▪ IS Salaries     |        | 75 000  | 82 500  | 90 750  |  |
| ▪ Upgrades        |        | 10 000  | 10 000  | 10 000  |  |
| Annual Prod.Costs |        | 100000  | 107500  | 115750  |  |
| (Present Value)   |        | 74 074  | 58 985  | 47 046  |  |
| Accumulated costs |        | 289 075 | 348 059 | 395 105 |  |
|                   |        |         |         |         |  |

B.

| Benefits                 | Year 0 | Year 1  | Year 2  | Year 3  |
|--------------------------|--------|---------|---------|---------|
| Improve Customer Service |        | 150 000 | 180 000 | 216 000 |
| Increase Productivity    |        | 150 000 | 187 500 | 234 375 |
| Total Benefits           |        | 300 000 | 367 500 | 450 375 |
| Present Value            |        | 222 223 | 201 647 | 183 052 |
| Accumulated Benefits     |        | 222 223 | 423 869 | 606 920 |
| Gain or Loss (NPV)       |        | 66 852  | 75 810  | 211 816 |
| Profitability Index (PI) | 0.99   |         |         |         |

Profitability index = 0.99, showing that it is not good investment because its index is less than one.

5. A.



B. Path 1 - A-B-C-D-F-H-J-K Length =  $3+2+2+3+4+7+2 = 23$

Path 2 - A-B-C-E-I-J-K Length =  $3+2+6+2+5+2 = 20$

C. Path 1 - A-B-C-D-F-H-J-K Length =  $3+2+2+3+4+7+2 = 23$

D.

